14 / KPI LIBRARY

# Supply Chain KPI Library

The analytical intelligence layer

**Purpose for the AI Assistant**
The canonical formulas behind every analytical answer. Derived-metric questions map to exactly one KPI here, so the assistant computes the same number the same way every time — no ad-hoc math.

**One KPI, one formula**
When a question needs a calculated number, the tool uses the formula on this page verbatim. This keeps answers reproducible and auditable across the whole org.

## 1. Inventory KPIs

### Inventory Value

| | |
|---|---|
| **Formula** | SUM(stock.available_amount) |
| **Tables** | stock |
| **Business meaning** | Financial worth of usable stock (PKR). Never quantity × price. |
| **Example question** | What is our inventory value? (group total is PKR 860,385,663) |

### Stock Coverage Days

| | |
|---|---|
| **Formula** | available_qty ÷ (issuance_3m ÷ 90) [avg daily consumption from last 90 days] |
| **Tables** | stock + issuance (or ab_items.stock_days_3m) |
| **Business meaning** | How many days current stock lasts at the recent consumption rate. |
| **Example question** | How long will item X last? |

### Hold Ratio

| | |
|---|---|
| **Formula** | SUM(hold_qty) ÷ NULLIF(SUM(stock_qty),0) |
| **Tables** | stock |
| **Business meaning** | Share of stock that is blocked — capital tied up and unusable. |
| **Example question** | How much of our stock is blocked? |

### Stock Health

| | |
|---|---|
| **Formula** | stock-days coverage ratio (ab_items.stock_health); higher = healthier |
| **Tables** | ab_items |

| | |
|---|---|
| **Business meaning** | Composite coverage signal. Low/zero flags reorder or critical. |
| **Example question** | Is item X healthy? |

### Critical Inventory

| | |
|---|---|
| **Formula** | COUNT(items in ab_items where list_type='critical') [currently 43] |
| **Tables** | ab_items |
| **Business meaning** | Items at high stockout risk; combine with lead time to prioritise. |
| **Example question** | How many items are critical? |

## 2. Procurement KPIs

### Supplier Delay (days)

| | |
|---|---|
| **Formula** | AVG(purchase_date – required_date) per supplier; positive = late |
| **Tables** | purchase |
| **Business meaning** | Average lateness of a supplier's deliveries against required dates. |
| **Example question** | Which suppliers are most delayed? |

### On-Time Delivery %

| | |
|---|---|
| **Formula** | 100 × COUNT(delay_days ≤ 0) ÷ COUNT(*) per supplier |
| **Tables** | purchase |
| **Business meaning** | Reliability: share of a supplier's lines delivered on/before required date. |
| **Example question** | What is supplier X's on-time rate? |

### Supplier Risk Score

| | |
|---|---|
| **Formula** | weighted( avg_delay, spend_share, critical/import item supply, on-time% ) |
| **Tables** | purchase + ab_items + import_shipment |
| **Business meaning** | Composite risk: late + high-spend + supplies critical/imported items = high risk. |
| **Example question** | Which suppliers create production risk? |

### Spend Concentration

| | |
|---|---|
| **Formula** | SUM(amount) of top-N suppliers ÷ SUM(amount) all |
| **Tables** | purchase |
| **Business meaning** | Dependence on a few suppliers (supply-continuity risk). |
| **Example question** | How concentrated is our spend? (194 active suppliers) |

## 3. Import KPIs

### Shipment Value Exposure (on water)

| | |
|---|---|
| **Formula** | SUM(total_value_pkr) WHERE current_status='In Transit' |
| **Tables** | import_shipment |
| **Business meaning** | Capital value currently at sea and not yet received. |
| **Example question** | How much value is on water? |

### ETA Slippage (days)

| | |
|---|---|
| **Formula** | eta (latest) – eta_1 (first ETA) |
| **Tables** | import_shipment |
| **Business meaning** | How much a shipment's arrival estimate has drifted — a delay signal. |
| **Example question** | Which shipments are slipping? |

### Country Dependency

| | |
|---|---|
| **Formula** | COUNT(shipments) per country ÷ total shipments |
| **Tables** | import_shipment |
| **Business meaning** | Concentration of imports by origin country (policy/customs risk). |
| **Example question** | How dependent are we on China? (160 of ongoing shipments) |

## 4. Logistics KPIs

### Transit Performance (days)

| | |
|---|---|
| **Formula** | AVG(actual_arrival – etd_karachi) or AVG(transit_time) |
| **Tables** | logistics_shipment |
| **Business meaning** | Average door-to-port transit; compare vs plan and by shipping line. |
| **Example question** | What is our average transit time? |

### Freight per kg

| | |
|---|---|
| **Formula** | SUM(total_shipping_cost) ÷ NULLIF(SUM(nw_kgs),0) |
| **Tables** | logistics_shipment |
| **Business meaning** | Shipping cost efficiency normalised by weight. |
| **Example question** | What is our freight cost per kg? |

### Documentation Completion %

| | |
|---|---|
| **Formula** | 100 × COUNT(docs_overall complete) ÷ COUNT(*) |
| **Tables** | logistics_shipment |
| **Business meaning** | Share of shipments with customs + customer + bank + BL docs complete. |

| | |
|---|---|
| **Example question** | How many shipments have complete docs? |

## 5. Consumption KPIs

### Consumption Value

| | |
|---|---|
| **Formula** | SUM(issuance.total_pric) |
| **Tables** | issuance |
| **Business meaning** | Value of material issued from stores. Never quantity × unit_price. |
| **Example question** | What did we consume? (window total PKR 3,522,817,955) |

### Department / Machine / Job Consumption

| | |
|---|---|
| **Formula** | SUM(total_pric) GROUP BY department \| machine \| job_number |
| **Tables** | issuance |
| **Business meaning** | Consumption attributed to an org unit or a production job (for costing). |
| **Example question** | How much did Production consume? |

### Consumption vs Purchase

| | |
|---|---|
| **Formula** | SUM(issuance.total_pric) vs SUM(purchase.amount) over same window |
| **Tables** | issuance + purchase |
| **Business meaning** | Are we buying roughly what we burn? Large gaps signal stockpiling or drawdown. |
| **Example question** | Are we over-buying? |

## 6. Usage note

- Composite KPIs (Supplier Risk, Stock Health) combine sources — the master agent fans out to the relevant department agents, then applies the weighting here.
- Every windowed KPI must report its date window alongside the number.
- Where postgres_schema.sql already exposes a view (e.g. v_supplier_delay), the tool calls the view rather than re-deriving the formula.